

Ans: D

Q5

When floating, net force on steel = 0. Therefore, Upthrust – weight = 0

$$V_{\text{submerged}} \times \rho_{\text{mercury}} \times g = V_{\text{steel}} \times \rho_{\text{steel}} \times g$$

Hence, fraction of steel submerged

$$= V_{\text{submerged}} / V_{\text{steel}} = \rho_{\text{steel}} / \rho_{\text{mercury}} = 8050 / 13600$$

$$= 0.59$$

Ans: C

Q6